

revenues in India will be approximately US \$17 billion by 2030.

Q How do you see 5G enabling immersive user experiences in the cloud gaming space?

A According to the Ericsson Consumer Lab, cloud gaming is amongst the top 10 services that Indian consumers are looking forward to in their 5G plans. Globally, 5G ready users are already spending more time on multiplayer mobile gaming, spending 1 hour more per week on multiplayer online games as compared to 4G users.

With a large youth population and increased smartphone penetration, India's cloud gaming ecosystem is expected to grow rapidly. Latency remains a big challenge especially in the cloud-gaming segment which impacts the gaming experience and discourages gaming companies to add ultra-high-definition graphics. Games are being developed with HD graphics for enabling an enriching and engaging viewing experience. The flavour of cloud-based gaming or live gaming have brought the significance of low latency to the forefront.

The unique characteristics of 5G, like high throughput and ultra-low latency, will enable an immersive mobile gaming experience in India. Alongside cloud and mobile edge computing, gamers will finally be able to experience seamless Virtual Reality (VR), Augmented Reality (AR) and Extended Reality (XR) capabilities that are free from lag and cables.

Q How will 5G enhance video consumption, augmented reality (AR) and Metaverse experiences in India?

A In this era of binge-watching, video consumption is the most significant traffic type generated by smartphone users. Approximately 7 in 10 of current smartphone users are looking forward to enhanced video streaming quality once they switch to 5G, and 360-degree live video streaming is amongst the top 10 services that Indian consumers wish for on their 5G plans.

Why is 700MHz band crucial for 5G in India

On the first day of India's 5G spectrum auction, the 700MHz band of spectrum received bids for over Rs 39,000 crore by all four Indian telecom operators, including Reliance Jio and Bharti Airtel, across all 22 circles of the country – that's close to 30-percent of all the bids submitted totalling over 1.4 lakh crore. According to experts, this 700MHz band is going to be crucial for wide-spread, uninterrupted 5G data transmission in the country. It's simple physics: radio waves with lower frequencies travel further through obstacles without degrading than waves with higher frequencies. Hence, the 700MHz band is being eyed by Indian 5G service providers with a lot of interest, because it is important to ensure 5G network services reach far and wide in every corner of the country without having to deploy a lot of signal towers (which will add to the infrastructure costs of telecom operators). Read more: <https://digit.in/5g700mhz>

Applications of augmented reality make the customer experience more efficient and comfortable. AR technology allows people to virtually model and try on items such as: clothing, houses, jewellery, makeup, cars, and furniture. So, we see more applications in Gaming, e-commerce and retail interior design. Eg consumers can visualize to see how the furniture /colour of the furniture items better fit their home layout etc. Likewise, AR lets potential buyers visualize properties by turning 2D blueprints into interactive 3D visualizations. These and more applications can be brought alive for Indian consumers at scale with 5G.

Coming to the metaverse, the definition itself is constantly evolving. Many elements of the metaverse have been around for years but depending on the

definition of what the metaverse is, or could be, the virtual worlds and immersive experiences that exist today represent a starting point for the metaverse.

According to a recent Ericsson ConsumerLab report, 5G users on average spend 1 hour more per week in metaverse-related services than 4G users. We also expect two hours of more video content to be consumed weekly on mobile devices, 1.5 hours of which will be on AR/VR glasses by 2025.

You see, 5G has already emerged as an important enabler for early adopters to embrace metaverse-related services, such as socializing, playing and buying digital items in interactive 3D virtual gaming platforms.

Q What will happen to older network technologies like 4G / 3G / 2G in the future?

A While 2G and 3G has been on a decline over the last few years, we expect 4G subscriptions to peak in 2024 at around 930 million and thereafter decline to an estimated 570 million by the year 2028. During the same timeline, as I mentioned earlier, 5G will be representing 53 percent of all mobile subscriptions in India. **d**

